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International AS and A-level

Physics (9630)
PH05 Physics in practice
Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH05 – JANUARY 2025

QUESTION 01

Students found this question very accessible, with most students scoring on all parts except **01.5**.

01.1 Most students were able to find the magnitude of $\boldsymbol{F}$. However, while most could apply Fleming's left hand rule, they did not appreciate that this gave the direction of the force on the wire, not the balance. An application of Newton's third law was required to realise the force on the balance was downwards.

01.2 Although most students could calculate the magnetic flux density, they did not give their answer to the minimum number of significant figures of the data used.

01.3 Many students were able to determine the absolute uncertainty in $\boldsymbol{\Delta m}$ or $\boldsymbol{l}$ but not both. The uncertainty in a digital reading for the mass, shown in Figure 1, is $\pm$ the least significant figure $\mathbf{0.01}$ g. Those who treated the zeroing of the scale as a measurement and therefore doubled the uncertainty also gained this mark.

The distance $\boldsymbol{l}$ was measured with a millimetre scale. Distances are measured with an analogue scale and therefore the precision is half the smallest division (0.5 mm). Both sides of the magnet must be measured to determine $\boldsymbol{l}$ so the total uncertainty is 1 mm.

The precision of an uncertainty should be given to 1 significant figure, so answers given to more than this did not score full marks.

01.4 Students were required to measure the induced emf from the graph and recognise that $\mathbf{40}$ ms was half the period. $\boldsymbol{B}$ could be obtained from $\boldsymbol{\varepsilon} = \boldsymbol{BAN\omega}\,\mathbf{sin}\,\boldsymbol{\omega t}$ by either using a value of $\boldsymbol{t}$ with a reading of $\boldsymbol{\varepsilon}$ at that time or equating the peak $\boldsymbol{\varepsilon}$ with $\boldsymbol{BAN\omega}$. Many students were able to score some marks but only a few scored all 3 marks.

01.5 Many students did not spot the anomalous data included in the table and therefore included it in their mean and uncertainty calculation, gaining 1 mark. To gain full marks, the anomalous reading of 0.089 had to be removed. The value of $\boldsymbol{B}$ and the uncertainty had to be given to the same number of decimal places as the data in **Table 1**.

01.6 Most students could use the data from **01.5** to determine the percentage uncertainty for their data. Very few students gave the detail required for the second mark. Some did not state that the uncertainty in method 2 was greater than 10%, with most stating that the uncertainty was 10%. Very few students gave an explicit comparison between the values and just stated that 'method 1 was smallest'; this comparison was required for the award of the second mark.

QUESTION 02

02.1 Most students were able to score 2 marks by choosing suitable scales and plotting the points in the correct position. Only a minority of students chose a difficult scale or did not mark the scale in an even way. A small number of students stopped the current scale at 100 which was less than the maximum value and so could not score the scale mark. When plotting points students should use a + or x; points marked with dots could not score both plotting points. The mark for the line was only given to smooth thin lines that were not dot-to-dot or formed from two or more joined lines.

02.2 The question asked students to explain how a graph of $\ln I$ against t could be used to determine C. Explain questions must be fully justified. While in this case this could mostly be done with algebra, some written explanation was expected. While many students correctly referred to the use of the gradient, many students used non-standard symbols for the gradient without defining them (e.g. k) and were unable to access the second mark.

Answers which did not provide an equation rearranged with C as the subject could not get the third mark. A significant majority gave the answer as gradient = $\frac{1}{RC}$ and stated that since R was known, C could be found; this was not enough for the award of the mark.

QUESTION 03

03.1 Students found this question very accessible with the majority of students scoring at least 3 marks.

The most accessible area was the processing of the results, with almost all able to describe how to determine the efficiency from the measurements taken. Where students were only partially able to address this area it was mostly due to answers which did not describe how the input power of the motor could be calculated from real measurements and just used the power of the motor.

Most students were able to describe some of the necessary measurements and some of the measuring devices but found it more challenging to describe all of the measurements and measuring devices. Most frequently students did not remember to state the measuring device used to measure time or the quantities measured by an ammeter and voltmeter. A significant minority just listed the measuring devices without stating how they were used or what they measured. A small minority did not attempt to measure the power using real devices (e.g. ammeter and voltmeter or joulemeter) and just stated the power should be measured or measured with a power meter, which was not accepted.

The least accessible area was how to perform the investigation accurately. Most students who attempted this area could partially address the area but only gave one technique so did not fully address the area by giving at least 2 techniques. Ideas including repeating and averaging, repeating and removing anomalies, the use of a datalogger and named sensors and using a slow motion video camera were accepted. The use of a set square to ensure the ruler was vertical was accepted but only if the answer indicated how the set square was used (in the text or in a diagram); just mentioning the use of a set square was not enough. It was not enough to state that parallax error should be avoided; a technique had to be described (e.g. ensuring the measurement was taken at eye level). Answers which referred to controlling the voltage to ensure it remained constant had to include a valid method to do this (e.g. adjusting the power pack or adjusting a variable resistor). The question referred to a constant supply voltage, so more than just mentioning that it had to be controlled was required.

03.2 Most students were able to score some marks for either stating that measurements with smaller increments or more measurements around the peak were required, but only a few made both points.

03.3 Only answers which linked the current mentioned in the question to the heat loss in the motor were able to score here. Answers which referred to friction or did not link the current to the heating did not score.

QUESTION 04

This question was very similar to 04 in June 2024. However, students found the use of the y –intercept in this paper more challenging than the use of the gradient in June.

04.1 Students were asked to draw minimum and maximum gradients. Most students did not appreciate that the lines had to pass through all of the error bars. Many students chose to link the first and the last error bars, creating lines which did not pass through all of the error bars. This approach was able to gain one mark this year. Lines which did not cross over between the first and last error bars were not regarded as attempts at maximum and minimum gradients and did not score.

A number of candidates started and stopped their lines with the first and last error bars. While this was awarded marks in 04.1 this year, it made determining the y–intercept in 04.2 much harder that it was intended to be. Students should draw lines on graphs across the whole range of the graph paper. This allows the mathematical relationship between the variables to be more easily determined. Some students initially drew lines that only covered the error bars and then realised they needed to determine the y–intercept in 04.2 and attempted to extend their line. This often resulted in lines which were not straight and could therefore not gain marks. Lines should be drawn in pencil and then erased and redrawn if they later need to be modified.

04.2 Students were required to realise that r was $-y$ intercept. This could be determined as the mean value of the y –intercepts from the maximum and minimum lines drawn. The uncertainty is given by half of the range of these values.

Students who did not extend their lines to the axis could gain marks by taking points from their lines and calculating the y –intercept, although this was much more demanding than the intended read-offs. In order for this to be a valid approach, the points had to come from the lines and so values from the edges of error bars were only credited if the line passed through those ends. If the lines were not extended to the y –axis and there was no evidence of a calculation of the y –intercept then this approach was limited to 1 mark.

The first mark required these values to be determined accurately from the graph. Error carried forward was allowed for the determining of r and its uncertainty from misreads. However, some students used values which bore no relationship to the lines on the graph – such answers were limited to 1 mark.

04.3 This answer required two parts. Around half the students were able to address one of the parts but only a very few were able to provide a full answer. Both parts of the answers were seen in students' responses, with slightly more giving the algebraic approach.

Students had to show how the equation $\boldsymbol{R} = \frac{\boldsymbol{k}}{\sqrt{\boldsymbol{C}}} - \boldsymbol{r}$ was consistent with the inverse square law. This required them to realise that the actual distance from the source to the detector was $\boldsymbol{R} + \boldsymbol{r}$ and to square this to give an inverse square equation. Students should realise this should be given in the form $\boldsymbol{C} = \frac{\boldsymbol{k}^2}{(\boldsymbol{R}+\boldsymbol{r})^2}$ but answers which grouped $(\boldsymbol{R} + \boldsymbol{r})$ and where both had been squared were condoned.

Students also had to show how the relationship given was consistent with the graph. An ideal answer would state that the graph showed a straight line with a negative y –intercept. Answers which referred to a linear relationship were condoned but answers which only referred to a straight line were not enough. Answers which incorrectly mentioned a directly proportional relationship did not get the mark.

QUESTION 05

05.1 Most students were able to use the resistivity equation to calculate the resistance of the aluminium. However, many did not realise that the cables were in parallel and tried to add the resistances. A significant minority of students were able to complete the calculation but gave their answer to 1 significant figure and so were only able to score 2 marks.

05.2 Most students were able to complete this successfully, although a very small minority calculated the power in the whole cable rather than the steel.

05.3 Most students were also able to complete this calculation successfully. Many students again gave their answer to 1 significant figure although this was condoned here, to avoid over-penalising students for the same mistake as in **05.1**.

05.4 This question was open-ended with a variety of valid approaches. The students were asked to compare the rate in the rise of the temperature between the aluminium and steel. A valid direct comparison between numerical data for the steel and aluminium had to be used for full marks.

The most straightforward and most common approach was to calculate $\frac{\Delta\theta}{\Delta t}$ for aluminium and compare this with the value for steel from **05.3**. Some students lost marks with this approach due to mixing up the data for 1 wire and all 6 wires of aluminium, most commonly using the resistance for 1 wire with the mass for 6 wires. This mistake only lost 1 mark if the answer was otherwise correct. The most common reason that answers did not gain full marks was the lack of an explicit numerical comparison. The number from **05.3** had to be restated with an explicit comparison using either symbols and words to gain full marks.

Other possible approaches used more rarely involved comparing the ratio of the powers in the aluminium and steel with the ratio of the heat capacities (mass × specific heat capacities) of the aluminium and steel wires. Very few answers were able to gain full marks using this route due to either not evaluating and comparing the ratios explicitly or by only comparing the specific heat capacities and ignoring the differences in mass. As with the other route, some responses mixed up the data for 1 wire and 6 wires of aluminium.

Two marks could be awarded for students who took a qualitative approach but most students who took this approach did not identify that the aluminium and steel had the same pd across them and were only able to score 1 mark.

This question was demanding with only around two thirds of students gaining at least 1 mark, and only around a quarter of students achieving full marks.

05.5 Only around a quarter of students were able to express the idea that the rate of heat transfers to the surroundings was the same as the rate of heating. Many answers did not get the mark because they compared the energy without reference to time or rate.

05.6 This question also provided challenging as most answers only referred to the wires having the same extension and original length, without giving a reason for why the cables have the same extension. To gain this mark, students had to explain why the extensions were the same with the idea that the wires were joined to the other wires in the cable. It was not enough to state that wires started and ended in the same place.

05.7 As a ‘show that’ question it was expected that for full marks students had to show equations and substitutions leading to the answer of 50% or 0.5. Many students assumed relationships that were not justified or did not show full substitutions and could not gain full marks.

05.8 This was a relatively straightforward calculation in principle, with most students accessing at least some of the marks and over half achieving full marks. Students could approach this problem by considering one wire of either steel or aluminium, all 6 wires of aluminium or the whole cable. Most marks were lost by mixing up the data between the 1 wire and 6 wires of aluminium. Most students who used the whole cable approach were unable to work out the effective Young Modulus of the cable, ignoring the effect of the different cross sectional areas due to the 6 aluminium wires, and merely added the two Young Modulus values together. They were therefore unable to access all of the marks. The majority of students who attempted the question using the steel wire successfully gained all 3 marks.

05.9 The calculation of the moment proved to be very accessible with most students gaining full marks. Only a small minority were unable to resolve the force or deal with the angle successfully.

05.10 This question proved challenging with only around one third scoring. Any reasonable weather condition backed by correct physics was rewarded, however small the effect. However, as the question asked for an explanation of how the moment changes, this required a statement about whether the moment would increase or decrease. Many marks were lost due to vague answers about the moment changing or stated the moment could increase or decrease. A few otherwise detailed answers did not gain the mark because they did not refer to the effect of their change on the tension in the cable, e.g. referring to the weight increase of snow without stating that this increased the tension in the cable.

Answers which discussed the effect of changes in temperature were approached in a few ways. While most answers correctly referred to the expansion of the cable with heating (or contraction with cooling), linking this to the effect on the moment was more challenging. The most straightforward way was to link this to an increase in the angle due to sagging when heated, decreasing the horizontal component of tension. However, alternatives which referred to a change in tension were condoned, although they often relied on incorrect physics beyond the scope of the specification, provided the physics on the specification was applied correctly.

QUESTION 06

06.1 Most students were able to state the meaning of coherence although a minority did only refer to the same frequency or a constant phase difference and not both. A reference to the waves being in phase was not condoned for a constant phase difference; only a small minority of answers used this phrase.

06.2 This question proved challenging for many students. A large number of students restated the definition of coherence from **06.1** without saying why the waves had the same frequency or a constant phase difference. Around half of the students were able to explain that the light coming from the same source was why they were coherent. The mark was awarded even if the student wrongly identified **A** as the source.

06.3 While nearly all students knew that a dark fringe was produced, this had to be accompanied by some valid explanation to get a mark. This proved challenging for many students. Ideally students would identify that the path difference was an integer number of wavelengths and therefore, without accounting for the phase shift on reflection, the waves would be in phase, with the phase change on reflection producing a dark fringe. However, an answer reflecting this detail was very rarely seen. In most cases students identified the overall phase shift or an effective path difference and identified this with a dark fringe. This was condoned in this series. However, it is expected that students would explicitly identify an integer or whole number of wavelengths rather than just stating a path difference of $\boldsymbol{48\lambda}$.

Answers which mixed up the phase difference and path difference or which did not reflect that the path difference was $\boldsymbol{2t}$ (rather than $\boldsymbol{t}$) were able to gain 1 mark.

06.4 Students were asked to show that an expression was valid using similar triangles. Therefore, full marks were only awarded to answers which started with an equation that could reasonably be expressed based on similar triangles. It was expected that answers would start with equal ratios based on side lengths. Good answers explicitly drew these triangles and labelled their sides, although this was not required for full marks. As a ‘show that’, it was expected for full marks that the use of $\boldsymbol{t}$ as one of the side lengths would be used along with $\boldsymbol{24\lambda = t}$ in some form. Answers which started with a side length of $\boldsymbol{24\lambda}$ had to justify this, usually with a diagram to gain both marks. Answers which tried to work backwards from the equation given did not score marks.

06.5 This question was accessible to most students. The small number of students who scored 1 mark either truncated the final answer rather than rounding or used

$\boldsymbol{3.0 \times 10^{8}}\ \mathrm{m\,s^{-1}}$ for the speed of light in air rather than the value given in the question.

06.6 Around three-quarters of students correctly identified that the wavelength decreases and that more fringes would be produced but only around a quarter were able to justify this satisfactorily. Most students who gained the second mark either referred to the fringe spacing or width decreasing, or used the equation from $\boldsymbol{h = H - \frac{n\lambda d}{x}}$ by modifying the equation given in **06.4**. Answers which referred to either the double slit or diffraction grating equation were not awarded this mark, as this situation was neither the double slit nor a diffraction grating. This approach did not prevent the award of 1 mark for correctly identifying the change in wavelength.

QUESTION 07

The calculations in **07.1**, **07.2**, **07.3** and **07.5** were referred to as estimates and therefore answers given to 1 significant figure were condoned, although it would have been preferred if students had given their answers to 2 significant figures. Rounding errors were penalised in these questions.

07.1 This question was accessible to the majority of students, with only a small minority losing marks due to errors involving powers of ten.

07.2 This question was generally accessible to most students. Errors related to powers of ten were more common than in **07.1**.

07.3 The question was also accessible, although again marks were lost due to errors in powers of ten. A minority of candidates did divide the percentage by 100, possibly mistaking the percentage change for a fractional change due to the small number involved.

07.4 This question proved very challenging for most students, with only around a quarter gaining a mark. Many students misinterpreted the question as only requiring a link between temperature and expansion in the graph. To answer the question students were required to explain why there was a larger rise in temperature in the top layers compared to the deeper layers. This could be answered in terms of the hotter less dense water floating to the surface (rather than mixing with the deeper layers), the sunlight not reaching the deeper layers or the poor thermal conduction of the water. Answers which referred to insulating the deeper layers or poor heat transfer without a reference to conduction did not provide the detail required. The few answers that went beyond the specification and referred to convection were able to gain the mark from the first approach about density.

07.5 Nearly all students were able to suggest an acceptable mean temperature for the Earth's surface and convert it to kelvin. However, some temperatures were near the edge of the accepted range and probably reflected a mean local temperature more than a mean global temperature. Most students were able to use this temperature with the provided equation. However, only around a third of students achieved full marks due to the required unit. While most students correctly identified the unit as metre kelvin, it was common to see lower case k used for kelvin and in a few cases an upper case M used for metres; neither got the mark.

07.6 Most students calculated the kinetic energy of a gas molecule and the energy of a photon. To achieve the third mark, students needed to calculate the ratio between the two and provide a comment on it, rather than simply stating which one was larger.

07.7 This question was challenging to most students with only around a third gaining at least a mark and small minority gaining two. Most answers were vague and not detailed enough. Students could refer to the emitted photons being reabsorbed by other greenhouse gases, but not just air particles. Often answers that came close either directly referred to the photons from the ground, rather than those emitted from the carbon dioxide, or were ambiguous about which photons were being referred to. The most common answer that gained marks was to refer to some of the emitted photons being directed towards the ground. Answers that referred to reflections back towards the ground from clouds or other named particles in the atmosphere also gained credit.

In general a lot of answers that did not gain credit either repeated the information in the questions or mentioned the greenhouse effect in general terms.

